

July 14, Monday

Experiment 1: Plasmid Extraction

Experiment Purpose: Extraction of plasmid

1. Preheat EB

- ddH₂O+ EB(preheated to 55°C)

2. Bacterial Pellet Processing

- Centrifuge(10,000rpm, 1min) 2 mL bacterial solution, discard supernatant.
- Add 250 µL solution 1, vortex to resuspend.
- Add 250 µL LB solution 2, invert tube 8 times, 30 s standing.

3. Neutralization

- Add 350 µL solution 3, mix by inverting 8 times.

4. Centrifugation

- Centrifuge at 10,000rpm for 10 min, transfer supernatant to spin column.
- Centrifuge column at 10,000rpm for 1 min, discard filtrate.

5. Wash

- Add 600 µL W2, centrifuge at 10,000rpm for 20s, discard filtrate.
- Add 500 µL W2, centrifuge at 10,000rpm for 20s, discard filtrate.

6. Centrifugation

- Leave the open column stand for 5min in order to let the ethanol evaporate.
- Place the spin column back into the collection tube, and centrifuge at 10,000rpm for 2 minutes.

6. Elution

- Add 80 µL TE, left stand for 1min, centrifuge at 10,000rpm for 1 min.

5. Result(edit)

Sample	Concentration (ng/ μ L)
plasmid	67.9

Experiment 2: PCR

Experimental Purpose: Amplification of the target gene

1. PCR Setup (50 μ L system)

- In a PCR tube, add 1 μ L of your template.
- Each of the Primer-F and Primer-R add 1 μ L.
- Add 25 μ L of enzyme(Vazyme phanta UniFi) to the mixture.
- Add 22 μ L of ddH₂O

2. PCR Setting

- Put your PCR tube into a thermal cycler.
 - Heat at 98°C for 40s to denature DNA.
 - Decrease the temperature to 58°C for 10s.
 - Heat to 72°C for 1min.
 - Repeat the above three steps 33 times.
 - Perform a final extension at 72°C for 5 min to ensure complete extension of all DNA fragments.
 - Store the PCR product at 4°C.
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Experiment 3: Agarose Gel Electrophoresis

Experimental Purpose: Verify the successful amplification of gene by PCR

1. Prepare Agarose Gel

- Weigh 0.45 g of agarose and add 30 mL of TAE buffer.

2. Dissolve Agarose

- Heat the mixture in a microwave oven at medium-high heat for 1 minute.

3. Cool Mixture

- Carefully remove the heated mixture using thick gloves and let it cool down.

4. Stain Gel

- Add 2.5 μL of 4s Green Plus and mix until a faint pink color is seen.

5. Pour Gel

- Pour the mixture into the electrophoresis gel mold and let it cool and solidify.

6. Load PCR Product

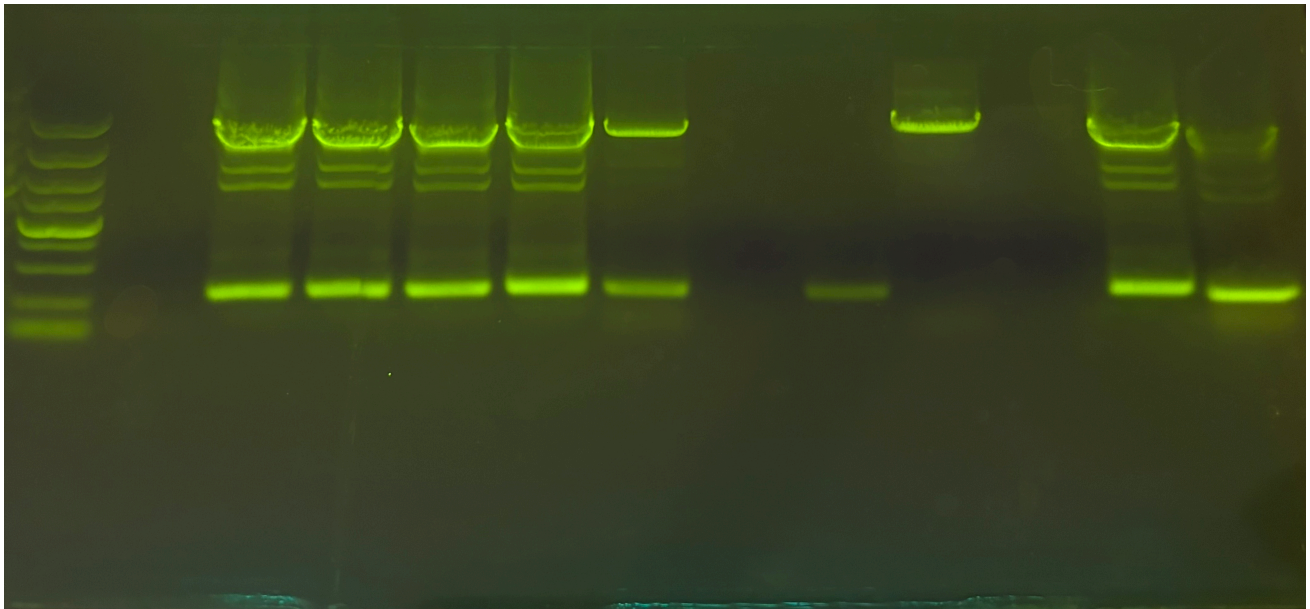
- Add 50 μL of PCR product or 10 μL of DNA marker to the wells of the agarose gel.

7. Run Electrophoresis

- Cover the gel with the electrophoresis lid and apply 120 V for 30 minutes.

8. PCR Results

- Succeed.



The one on the left side is the DNA Marker

The others on the right are PCR products